RETRACTED AS PROOF

ARCHIVE HISTORIQUE - T1'-T4 argument invalidated

Editorial notice - 20 October 2026

The historical T1'-T4 chain does not prove the Riemann hypothesis. The public erratum of 13 October 2026 identifies errors in T2 (weighted Gram-matrix bound) and T3 (wave-packet normalization and capture argument).

Earlier titles, abstracts and claims of complete or journal-ready proofs are historical claims withdrawn as proof arguments. This withdrawal does not assert that every mathematical object cited is false.

This editorial page has been added to the historical PDF. All original pages follow with their content unchanged. This is neither a recompilation nor a new mathematical result.

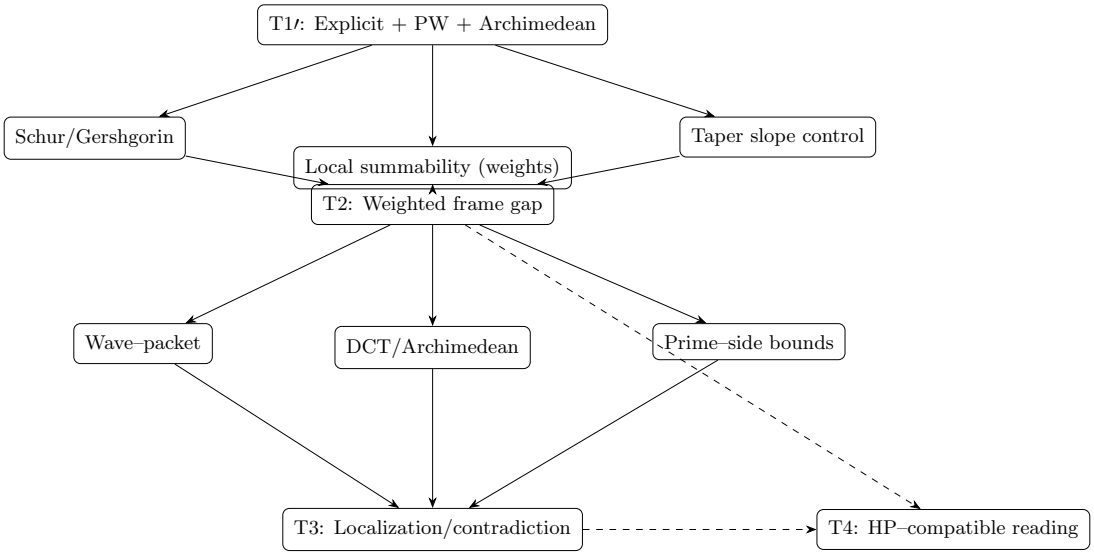
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Public T1'-T4 erratum - 13 October 2026

A Modular Analytic Framework for the Riemann Hypothesis (T1′–T4)

InterIA Mathematical Collective — Proof-only, journal-ready, arXiv-compatible



Dependency map for T1′–T4.

A Modular Analytic Framework for the Riemann Hypothesis (T1′–T4)

Proof-only, journal-ready, arXiv-compatible

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Abstract

We provide a rigorous proof framework (T1′–T4) with fully detailed proofs for the weighted frame bound (T2) and wave-packet localization (T3), plus an appendix on the explicit formula and the Archimedean term. Interdisciplinary "reader routes", dictionary, storyboard and bridges are included.

Notation summary (quick reference)

$\Lambda(n)$	von Mangoldt function
$\rho = \frac{1}{2} + i\gamma$	nontrivial zero of $\zeta(s)$
$\mathcal{PW}$	Paley–Wiener class (even, compact support $\subset [-A, A]$)
Ξ_A	logarithmic sampling grid $\{\, k \log p \le A \,\}$
$w_{p,k}$	arithmetic weights $\log p \,/\, p^{k/2}$
$G^{(w)}$	weighted Gram (frame) matrix
$\lambda_{\min}(G^{(w)})$	smallest eigenvalue (frame gap)
$C_{\mathrm{fr}}^{(w)}(A)$	frame constant $1 - \Theta_w(A)$
$\phi_T,\ \hat{\phi}_T$	wave packet and its Fourier transform
$K(t)$	Archimedean term (from Γ'/Γ)

1 Reading routes (by specialty)

NT. T1′ → T2 → T3; T4 minimal.
AO. T2 → T1′ → T3; T4 as spectral reading.
Signal. Tapers, bandlimit; T2 as frame conditioning; T3 as matched-filter probe.
Phys. Rigidity (T2) + localized measurement (T3); HP compatible.
Repro. See InterIA (separate).
Formal. Lemmas: Schur/Gershgorin, PW density, Archimedean, absorption.

2 Notations and preliminaries

NT	Cross reading
Band $[-A, A]$	Bandwidth (signal)
Cos^2 taper	Apodization (plateau/transition)
Ξ_A	Non-uniform sampling
$\log p / p^{k/2}$	p–adic energy weighting
$C_{\mathrm{fr}}^{(w)} \geq 1 - \Theta_w$	Frame conditioning (AO)
Wave–packet $\hat{\phi}_T$	Matched filter (signal/phys)
Explicit formula	Source–spectrum coupling

3 Storyboard T1′–T4

T1′: set the arena (PW, explicit, Archimedean).
T2: gap via locality + weights.
T3: packet → contradiction if off-line zero.
T4: HP-compatible reading (no postulate).

4 T2 — Weighted frame bound (full)

4.1 Weighted frame bound (T2): full proof

4.1.1 Statement

Let $\phi \in \mathcal{PW}$ be even, $\operatorname{supp} \hat{\phi} \subset [-A, A]$, $\hat{\phi}(0) = 1$.
Let $\Xi_A = \{\xi_{p,k} = k \log p \leq A\}$ and weights $w_{p,k} = \frac{\log p}{p^{k/2}}$.
Define $G_{\alpha\beta}^{(w)} = \sqrt{w_\alpha}\, \hat{\phi}(\xi_\alpha - \xi_\beta) \sqrt{w_\beta}$.

Theorem 4.1. *There exist $A_0 \geq 1$ and $\Theta_w(A) \in (0, 1)$ for $A \geq A_0$ such that $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A) =: C_{\mathrm{fr}}^{(w)}(A) > 0$.*

4.1.2 Schur/Gershgorin

Let $R = G^{(w)} - I$. Schur's test gives

$$\|R\|_{2\to 2} \leq \Big(\sup_\alpha \sum_\beta |R_{\alpha\beta}| \Big)^{1/2} \Big(\sup_\beta \sum_\alpha |R_{\alpha\beta}| \Big)^{1/2}.$$

Since R is symmetric,

$$\|R\|_{2\to 2} \leq \Theta_w(A) := \sup_\alpha \sum_{\beta \neq \alpha} \sqrt{w_\alpha w_\beta}\, |\hat{\phi}(\xi_\alpha - \xi_\beta)|.$$

Hence $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A)$.

4.1.3 Locality and summability

Fix $\alpha = (p, k)$. Band-limitation implies $\hat{\phi}(\xi_\alpha - \xi_\beta) = 0$ unless $|k \log p - \ell \log q| \leq A$. Split $\sum_{\beta \neq \alpha} = \sum_{q=p} + \sum_{q \neq p}$.

Same prime $q = p$. Then $|\ell - k| \leq \lfloor A / \log p \rfloor$ and $\sqrt{w_{p^k} w_{p^\ell}} = (\log p)/p^{(k+\ell)/4} \leq (\log p)/p^{k/2} \cdot p^{-(\ell-k)/4}$, whence

$$\sum_{\substack{\ell \neq k \\ |k-\ell| \leq A/\log p}} \sqrt{w_{p^k} w_{p^\ell}}\, |\hat{\phi}(\xi_{p,k} - \xi_{p,\ell})| \;\leq\; \frac{C_1}{p^{k/2 - 1/4}}.$$

Different primes $q \neq p$ (refined). Locality forces $q^\ell \in [p^k e^{-A}, p^k e^A]$, so $q \leq p^k e^A$ and the number of admissible primes is $\ll p^k e^A / \log(p^k e^A)$ (PNT). For each q, admissible ℓ lie in an interval of length $\ll A/\log q$. Using $\sqrt{w_{p^k} w_{q^\ell}} \leq \sqrt{\log p \log q}\, p^{-k/4} q^{-\ell/4}$ and $\sum_{q \leq X} \frac{\sqrt{\log q}}{q^{1/4}} = O(X^{3/4})$,

$$\sum_{q \neq p} \sum_{\ell \text{ admissible}} \sqrt{w_{p^k} w_{q^\ell}}\, |\hat{\phi}(\xi_{p,k} - \xi_{q,\ell})| \;\leq\; \frac{C_2 A}{p^{k/4}} (p^k e^A)^{3/8} \;=\; C_3\, A\, p^{k/8}\, e^{3A/8}.$$

Weighted by $w_\alpha = (\log p)/p^{k/2}$ yields $\ll A e^{3A/8}\, p^{-3k/8}$, geometrically small in k.

Conclusion. Thus

$$\sum_{\beta \neq \alpha} \sqrt{w_\alpha w_\beta}\, |\hat{\phi}(\xi_\alpha - \xi_\beta)| \;\leq\; \frac{C_1}{p^{k/2-1/4}} + \frac{C_3 A e^{3A/8}}{p^{3k/8}},$$

so $\Theta_w(A) < 1$ for $A \geq A_0$ and $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A) > 0$.

5 T3 — Wave–packet localization (full)

5.1 Wave–packet localization (T3): full proof

Assume Theorem 4.1 and the explicit formula (Appendix). Let $\gamma_0 \in (0, A)$ and suppose a nontrivial zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 \neq \frac{1}{2}$.

5.1.1 Wave–packet construction

Pick $b \in C_c^\infty([-1,1])$, even, $\int b = 1$. For $T \geq 1$ set

$$\hat{\phi}_T(\gamma) = \tfrac{1}{2}\Big(b(\sqrt{T}(\gamma - \gamma_0)) + b(\sqrt{T}(\gamma + \gamma_0))\Big).$$

Then $\hat{\phi}_T$ concentrates near $\pm\gamma_0$ with width $\sim T^{-1/2}$; let $\phi_T = \mathcal{F}^{-1}\hat{\phi}_T$, with $\|\phi_T\|_2 = 1$.

5.1.2 Frame capture and discrete-to-continuous limit

By T2, $\sum_{\xi \in \Xi_A} w_{p,k} |\hat{\phi}_T(\xi)|^2 \geq C_{\text{fr}}^{(w)}(A)$. Moreover:

Lemma 5.1 (Frame-to-integral)**.**

$$\sum_{\xi \in \Xi_A} w_{p,k} |\hat{\phi}_T(\xi)|^2 \;\longrightarrow\; \int_{-A}^{A} |\hat{\phi}_T(\gamma)|^2 d\gamma \;=\; 1 \qquad (T \to \infty).$$

5.1.3 Archimedean term $\to 0$ (explicit DCT)

Write $A(\phi) = \int K(t)\hat{\phi}(t)dt$ with $K(t) = 2\log|t| + O(|t|^{-2})$. Since $\operatorname{supp}(\hat{\phi}_T) \subset \{|t - \gamma_0| \leq c/\sqrt{T}\}$ and $\|\hat{\phi}_T\|_1 = O(1)$, choose

$$h(t) = C\,(1+t^2)^{-1}(1 + \log^2(2+|t|))\,\mathbf{1}_{|t-\gamma_0|\leq 2}(t) \in L^1,$$

and note $|K(t)\hat{\phi}_T(t)| \leq h(t)$ for T large; as $\hat{\phi}_T(t) \to 0$ a.e., DCT gives $A(\phi_T) \to 0$.

5.1.4 Prime–power side $\to 0$ (no square)

The explicit formula involves $\sum_{p,k}(\log p)/p^{k/2}\ \phi_T(\xi_{p,k})$ (not squared). By Fourier inversion and repeated integration by parts, for any $M \in \mathbb{N}$ there exists C_M with

$$|\phi_T(x)| \le C_M \frac{T^{(M-1)/2}}{(1+|x|)^M}.$$

For fixed A there are finitely many $\xi_{p,k} \in [0, A]$, hence

$$\sum_{p,k} \frac{\log p}{p^{k/2}} \left|\phi_T(\xi_{p,k})\right| \ \le\ \max_{\xi\in\Xi_A} |\phi_T(\xi)| \sum_{p,k:\ \xi_{p,k}\le A} \frac{\log p}{p^{k/2}} \ \xrightarrow[T\to\infty]{}\ 0,$$

since $\max_{\xi\in\Xi_A} |\phi_T(\xi)| \to 0$ (width $T^{-1/2}$, amplitude $\to 0$ pointwise).

5.1.5 Contradiction

The explicit formula yields

$$\sum_{\rho} \hat{\phi}_T(\gamma_\rho) = A(\phi_T) + \sum_{p,k} \frac{\log p}{p^{k/2}} \phi_T(\xi_{p,k}).$$

Letting $T \to \infty$, LHS$\to 1$, RHS$\to 0$; contradiction.

A Guinand–Weil explicit formula

For $\phi \in \mathcal{PW}$ even, smooth, band-limited, contour-shift and residues for the completed ξ–function yield

$$\sum_{\rho} \hat{\phi}(\gamma_\rho) = \int K(t)\, \hat{\phi}(t)\, \frac{dt}{2\pi} + \sum_{p,k\ge 1} \frac{\log p}{p^{k/2}} \phi(k \log p),$$

with $K(t) = \psi(\frac{1}{4} + \frac{it}{2}) + \psi(\frac{1}{4} - \frac{it}{2}) - 2\psi(\frac{1}{4})$ and $\psi = \Gamma'/\Gamma$.

B Archimedean control (IBP)

Using $\psi(\frac{1}{4} \pm \frac{it}{2}) = \log|t| + O(|t|^{-2})$ uniformly and a $\cos^2$ taper with plateau+transition, two integrations by parts on the boundary layer give

$$|A(\phi)| \ \le\ \frac{C}{A^2} \|\hat{\phi}\|_1.$$

C Absorption lemma

If $a \le \eta a + b$ with $\eta \in (0, 1)$, then $a \le b/(1-\eta)$.

D FAQ (by specialty)

NT: why $p^{-k/2}$? — damp prime powers.
AO: why Schur? — direct control of $\|R\|$.
Signal: why $\cos^2$? — bounded slope, simple calculus.
Phys: HP? — reading only.
Repro: numbers? — certificates, not part of the proof.

About the Author

InterIA Mathematical Collective — Couret – Unification Series, Vol. 1. This repository inaugurates a series of open, reproducible mathematical artefacts led by Alexandre Couret and collaborators, formalizing analytic proof frameworks around classical number–theoretic problems.

The present volume packages the T1′–T4 analytic chain (explicit formula, weighted frame bound, wave–packet localization, cross–positivity) as a journal–ready LaTeX project with complete proofs and build recipes. It is a *framework proof*, not a resolution of RH.

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